EXCEL ACADEMY

Class 10: Control and Coordination

1. Introduction to Control and Coordination

- All living organisms respond and react to changes in the environment, and these changes are called **stimuli**.
- **Control and coordination** is the process by which different parts of an organism's body work together in a systematic way to respond to these stimuli, ensuring the proper functioning of the organism.
- In animals, this is achieved by the **nervous system** and the **endocrine system** (hormonal system).
- In plants, coordination is primarily achieved through chemical substances known as **plant hormones** or phytohormones.

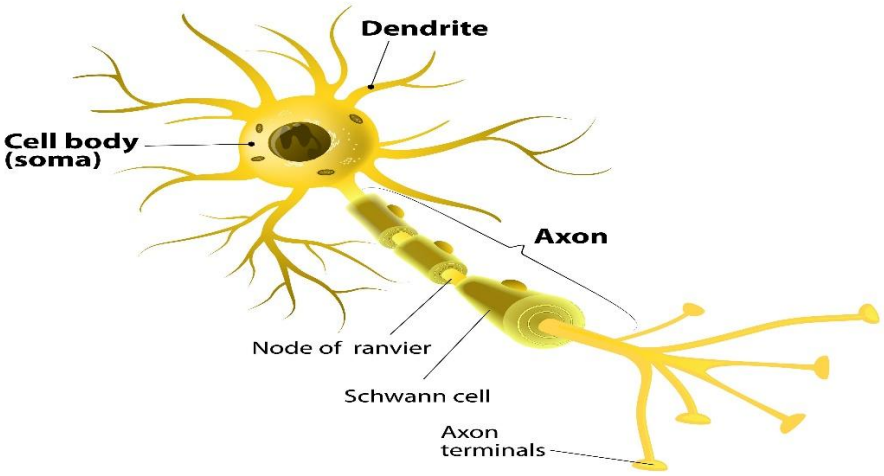
---

2. Nervous System in Animals

The nervous system is a complex network of nerves and cells that carry messages to and from the brain and spinal cord to various parts of the body.

Neuron: The Structural and Functional Unit

- The **neuron** (or nerve cell) is the basic unit of the nervous system.



- It has three main parts:
  - **Dendrite:** These are branched, tree-like structures that receive information from other neurons and pass it on to the cell body.
  - **Cell Body (Cyton):** This is the main part of the neuron, containing the nucleus and other organelles. It processes the information received.
  - **Axon:** This is a long, slender projection that carries the nerve impulse away from the cell body to the next neuron. It is often covered by a protective **myelin sheath**.

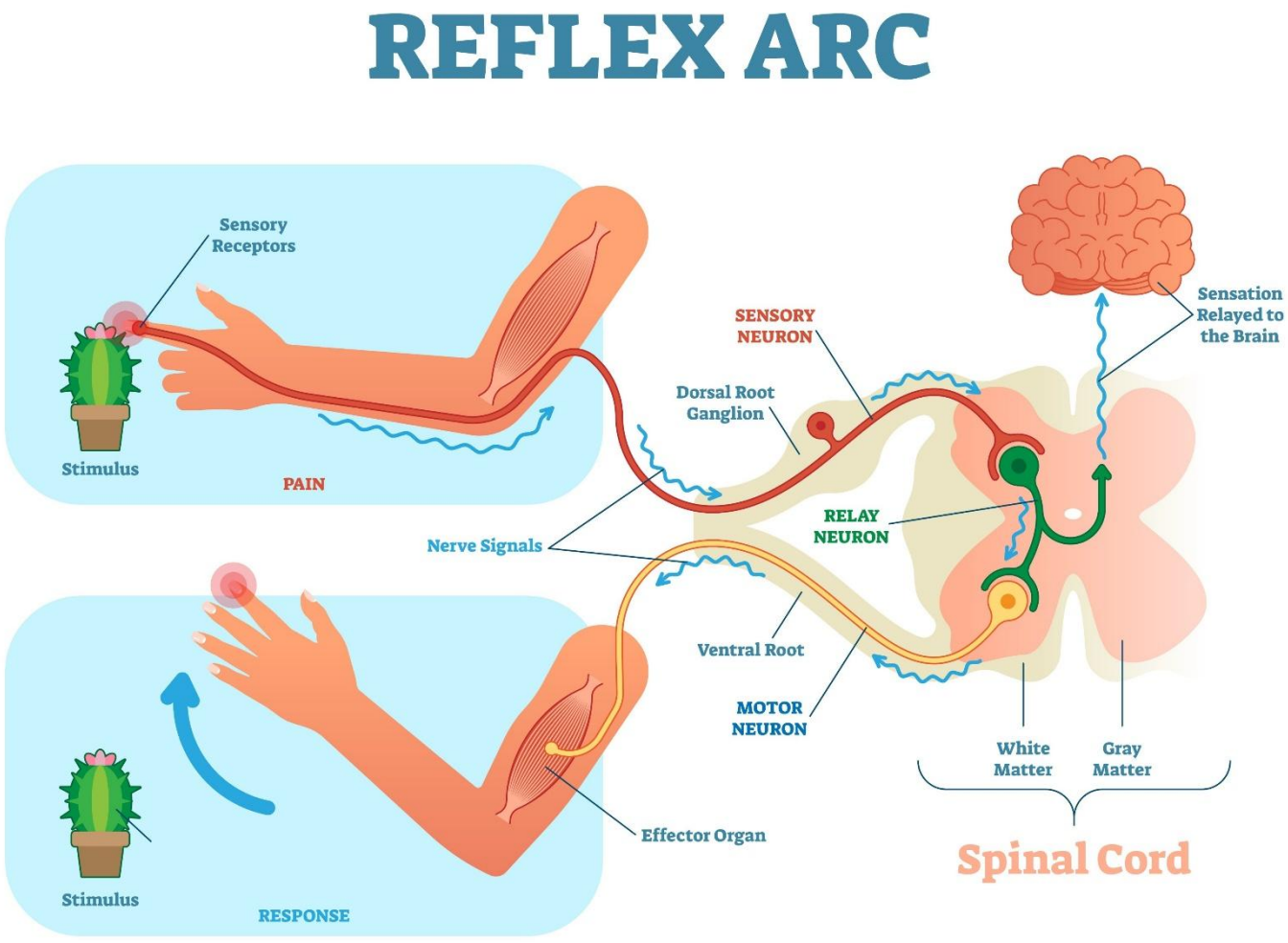
- The nerve impulse is an electrical signal that travels along the neuron.

The Synapse

- A **synapse** is the small gap between the axon terminal of one neuron and the dendrite of the next neuron.
- When an electrical impulse reaches the end of an axon, it triggers the release of chemical substances called **neurotransmitters** into the synapse.
- These chemicals diffuse across the gap and bind to receptors on the next neuron, starting a new electrical impulse.

Reflex Action and Reflex Arc

- A **reflex action** is a sudden, involuntary, and rapid response to a stimulus, which is not under conscious control. For example, immediately withdrawing your hand after touching a hot object.
- The pathway that a nerve impulse takes during a reflex action is called the **reflex arc**.



Licensed by Google

- The typical pathway of a reflex arc is: **Receptor** (e.g., skin) → **Sensory Neuron** → **Spinal Cord** (relay neuron) → **Motor Neuron** → **Effector** (e.g., muscle).
- Reflex actions are crucial for survival as they provide a quick response to harmful situations, bypassing the time taken for conscious thought in the brain.

Human Nervous System

The human nervous system is divided into two main parts:

A. Central Nervous System (CNS) The CNS is the main control center of the body and consists of the brain and the spinal cord.

- **Brain:**
    - The brain is protected by a bony box called the **cranium** (skull) and is covered by three membranes called **meninges**. The fluid between the meninges, **cerebrospinal fluid**, protects it from mechanical shock.
    - **Forebrain:** This is the largest part of the brain, consisting mainly of the **cerebrum**. It is the main thinking part of the brain and controls voluntary actions, memory, intelligence, and processes sensory information (hearing, sight, smell).
    - **Midbrain:** It connects the forebrain and the hindbrain and controls involuntary actions and reflexes involving the head, neck, and eyes (e.g., changing pupil size).
    - **Hindbrain:** This part consists of the cerebellum, pons, and medulla.
        - **Cerebellum:** Controls posture, balance, and the precision of voluntary actions like walking or writing.
        - **Pons:** Regulates respiration.
        - **Medulla (or Medulla Oblongata):** Controls involuntary actions such as heartbeat, blood pressure, breathing, and swallowing.
- **Spinal Cord:**
    - This is a cylindrical bundle of nerves enclosed within the vertebral column (backbone).
    - Its primary function is to control reflex actions and to conduct nerve impulses from the body to the brain and from the brain to the body.

B. Peripheral Nervous System (PNS) The PNS consists of the nerves that extend from the CNS to all other parts of the body.

- It includes **cranial nerves** (arising from the brain) and **spinal nerves** (arising from the spinal cord).
- The PNS is further divided into the **somatic nervous system** (controls voluntary muscle actions) and the **autonomic nervous system** (controls involuntary actions of internal organs).

---

3. Coordination in Plants 🌱

Plants do not have a nervous system, but they use hormones to coordinate their growth and response to environmental stimuli.

Plant Movements

Plant movements can be categorized into two types:

- **Tropic Movements (Directional):** These are directional growth movements in response to a specific stimulus.
    - **Phototropism:** The movement of a plant part in response to light. Stems grow towards light (**positive phototropism**), while roots grow away from it (**negative phototropism**). This is caused by the hormone **auxin**.
    - **Geotropism:** The movement of a plant part in response to gravity. Roots grow downwards towards gravity (**positive geotropism**), while shoots grow upwards against gravity (**negative geotropism**).
    - **Hydrotropism:** The movement of a plant part in response to water. Roots grow towards water.
    - **Chemotropism:** The movement of a plant part in response to chemicals. An example is the growth of the pollen tube towards the ovule during fertilization.
    - **Thigmotropism:** The directional growth of a plant part in response to touch. For example, the coiling of tendrils around a support.
- **Nastic Movements (Non-directional):** These movements are independent of the direction of the stimulus.
    - An example is the folding of leaves in the "touch-me-not" plant (*Mimosa pudica*) when touched.
    - This rapid response is caused by a sudden loss of water from specialized structures called pulvini at the base of the leaves, leading to a change in **turgor pressure**.

Plant Hormones (Phytohormones)

- **Auxins:** Promote cell elongation, cell division, and are responsible for phototropism. They are synthesized at the tip of the shoot.
- **Gibberellins:** Promote stem elongation and help in breaking the dormancy of seeds and buds.
- **Cytokinins:** Promote cell division and are found in high concentrations in areas of rapid growth, like fruits and seeds.
- **Abscisic Acid (ABA):** This is a growth inhibitor. It promotes dormancy in seeds and buds and causes the wilting of leaves. It is also known as the "stress hormone."

---

4. Hormones in Animals (Endocrine System)

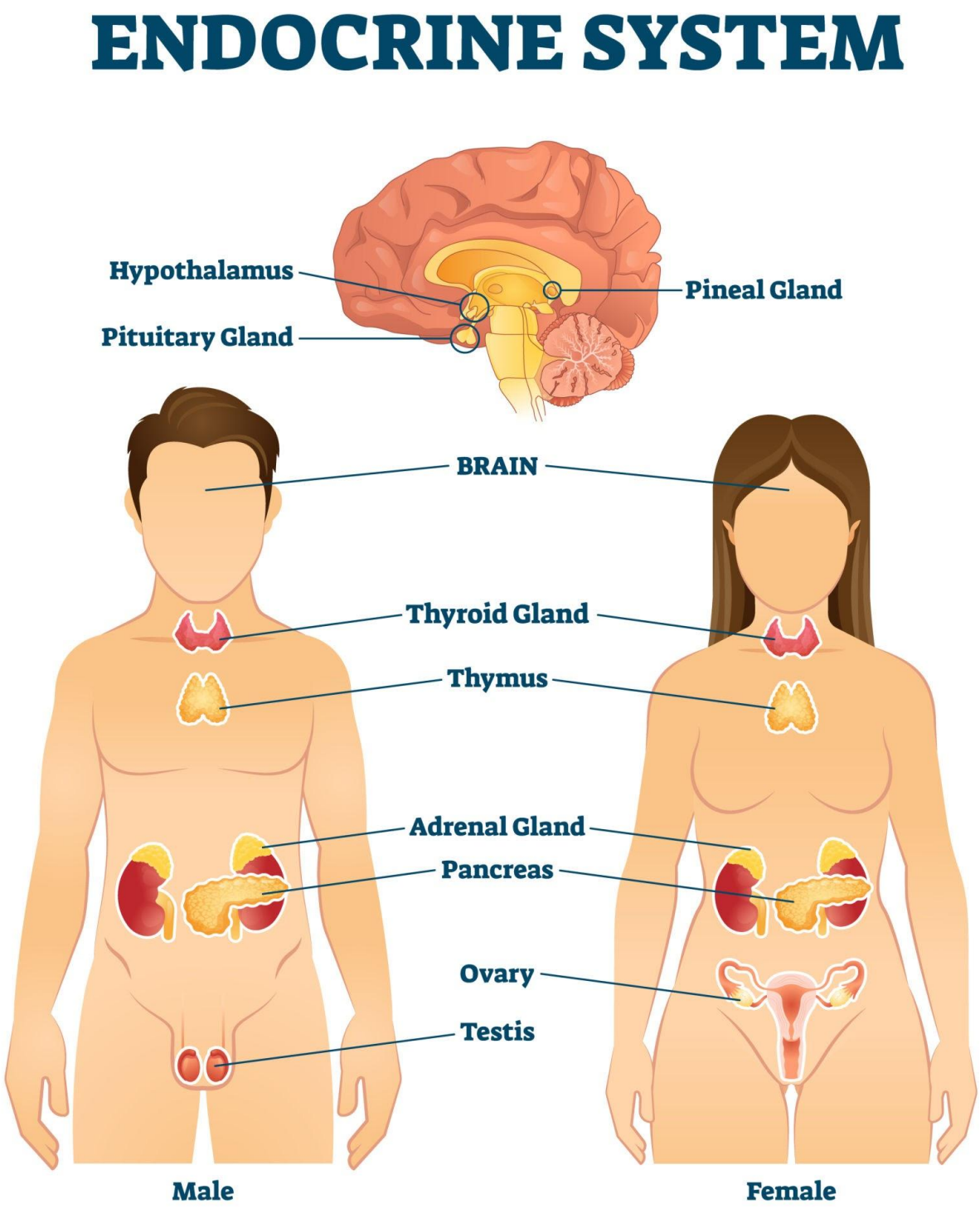
The endocrine system consists of glands that produce chemical messengers called hormones, which regulate various bodily functions.

Hormones and Endocrine Glands

- **Hormones** are chemical substances secreted by **endocrine glands** directly into the bloodstream.
- Endocrine glands are **ductless glands**, meaning they do not have tubes to release their secretions.

- Hormones travel through the blood to specific **target organs** or tissues to produce a particular effect.

Major Human Endocrine Glands and their Functions



Licensed by Google

- **Hypothalamus:** Located in the brain, it produces releasing hormones that control the secretions of the pituitary gland.

- **Pituitary Gland (Master Gland):** Located at the base of the brain, it controls the functioning of other endocrine glands. It secretes **Growth Hormone (GH)**, which regulates growth and development.
- **Thyroid Gland:** Located in the neck, it secretes **thyroxine**, which regulates the body's metabolism (rate of energy use). **Iodine** is essential for the synthesis of thyroxine; its deficiency can lead to a condition called **goitre**, characterized by a swollen neck.
- **Adrenal Glands:** Located on top of the kidneys, they secrete **adrenaline**. Adrenaline is the "fight or flight" hormone, preparing the body for emergency situations by increasing heart rate, breathing rate, and blood flow to muscles.
- **Pancreas:** Functions as both an exocrine and an endocrine gland. It secretes **insulin** and **glucagon**.
    - **Insulin** lowers the blood sugar level. Deficiency of insulin causes **diabetes mellitus**.
    - **Glucagon** raises the blood sugar level.
- **Testes (in males):** Produce **testosterone**, which is responsible for the development of male secondary sexual characteristics during puberty.
- **Ovaries (in females):** Produce **estrogen** and **progesterone**. Estrogen is responsible for female secondary sexual characteristics, and progesterone helps maintain pregnancy.

Feedback Mechanism

- The timing and amount of hormone released are regulated by a **feedback mechanism**.
- This mechanism ensures that hormones are secreted in precise quantities and at the right time.
- For example, if blood sugar levels rise, the pancreas secretes more insulin. As blood sugar falls, insulin secretion is reduced. This is an example of **negative feedback**.